

Skill Expansion Plan

Manthan Agent v7.9.12

A practical recommendation of which skills from 13 leading AI assistants can be wired into the SahuAI PRO-Lite chat app — bucketed by effort, with exact directive shapes, system-prompt snippets and an implementation roadmap.

Source catalog	AI Assistant Skills — Unified Report (Full Edition, 786 pages)
Catalog compiled	25 September 2026
Subject app	Manthan Agent v7.9.12 — SahuAI PRO-Lite (Beta)
Existing skill count	11 directive-driven skills (web, code, files, swarm, ...)
Recommended additions	16 quick-win (Sprint 1–4) + 12 heavier (Sprint 5–6)
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1. Executive Summary

The Unified Skill Report pulls together skill definitions from 13 AI assistants — Sarvam AI, ChatGPT, Grok, GLM (Z.ai), Mimo (Xiaomi/OpenClaw), MiniMax (MAVIS), DeepSeek, hy, Claude, Gemini, Meta AI (Muse Spark), Kimi (271 skills) and Manthan Agent itself — totalling well over 400 distinct skills.

Manthan Agent v7.9.12 (the SahuAI PRO-Lite web app, the same file as this document's companion `index_2.txt`) already ships with 11 directive-driven skills, a 23-engine search catalogue in 6 categories, a Pyodide + esm.sh sandbox, and 57+ models across 15 providers. The platform is mature; what it lacks is breadth in vision, voice, persistent memory, translation, and a handful of high-leverage integrations.

Key recommendations at a glance

READY	Bucket A — 16 skills implementable now Pure-LLM shims or direct browser calls. Drop into SAHU_SKILLS and the directive parser. Estimated effort: 4–8 days total.
HEAVY	Bucket B — 12 skills, larger glue Need a thin backend (OAuth relay, serverless Chromium) or are larger integrations (Email, GitHub, Notion, Obsidian). 1–3 weeks.
SKIP	Bucket C — not feasible browser-side Native macOS CLIs, telephony, job-runners, hardware bridges. Out of scope.
DUP	Bucket D — duplicates of existing 11 Same shape as skills you already ship; consolidate descriptions and borrow QA gates from Kimi/Mimo rather than ship a second copy.

Top 6 candidates by effort × value

#	New skill	Bucket	Effort	Why now
1	Translation	A	½ day	Pure LLM shim; multi-language is core to Sarvam/GLM/Kimi.
2	Humanizer	A	½ day	Removes AI tells from pasted text — high-frequency user need.
3	OCR	A	1 day	DeepSeek OCR / PaddleOCR models already in your model catalogue.
4	STT + TTS (wire voice)	A	1 day	<code>state.voiceConfig</code> exists but UI is gone; the engine was never re-hooked.
5	Persistent Memory	A	1 day	Mimo/Kimi have a full <code>MEMORY.md</code> + heartbeat spec; pure <code>localStorage</code> .
6	Image Generation	A	2 days	Browser-callable hosted models (FLUX, SDXL, Ideogram).

2. Current State of Manthan Agent

Manthan Agent v7.9.12 is a single-page browser application. Skills live as entries in the SAHU_SKILLS array (line ~11240 of index_2.txt). Each entry produces one block of system-prompt text via `buildSearchLoopPrompt()` and is matched against the model's output by a directive regex in the parser.

2.1 The 11 existing skills

#	Skill	Directive	Mechanism
1	Web Search	SEARCH[key]: q	23 engines in 6 categories
2	Web Scraping	READ: url	Full readable text extraction
3	Web Fetching	FETCH: url	Raw payload
4	Web Browsing	BROWSE: url	Summary + outbound links
5	Code Interpreter	```runpython / ```runjs	Pyodide + esm.sh
6	File Creation	FILE[ext]: name ... /FILE	txt/md/csv/json + xlsx/docx/pdf/pptx
7	Live Preview	PREVIEW: title + fence	html / react sandboxed apps
8	Skill Creator	MAKESKILL: ... /MAKESKILL	Custom skills persisted to Settings
9	MCP: Date & Time	DATE: [tz]	IANA timezone aware
10	MCP: Calculator	CALC: expr	Tokenizer + parser, no eval
11	Agent Swarm	```sahu_agents ...```	2–10 parallel workers, multiple rounds

2.2 Adjacent platform features already in place

- 15 providers, 57+ models (NVIDIA, Sarvam, Sambanova, HuggingFace, OpenRouter, Groq, Together, Cerebras, Fireworks, Routeway, Mistral, Gemini, Longcat, Ollama, Zenmusk, IONet).
- Vision-capable models auto-detected (Phi-3.5-Vision, Gemma-4, Llama-3.2-Vision, Kimi-K2, Mistral-Small-4, Qwen3-VL).
- DeepSeek OCR & PaddleOCR exposed as models (used as API, not yet as a directive).
- Custom search engines, custom categories, custom skill toggle order.
- File / image / camera / video / doc upload (txt, pdf, doc, docx, md, json, csv, js, py, html, css, png, jpg, gif, webp, svg, bmp, wav, mp3, webm, mp4).
- `state.voiceConfig` (TTS = `nvidia/magpie-tts-multilingual`, ASR = `openai/whisper-large-v3`) — UI removed but state kept.
- DOMPurify sanitization on all model output (XSS fix), Tailwind utility CSS precompiled and embedded, v7.9.13 bugfix pass complete.

3. Bucket A — Implementable Now

These 16 skills map cleanly onto your existing directive-parser envelope. Each adds one row to SAHU_SKILLS, one branch in buildSearchLoopPrompt(), and one regex in the directive extractor.

#	Skill	Directive shape	Source in PDF	Implementation hint
A1	OCR	OCR:	Kimi (deepseek-ocr/paddleocr), Mimo (openai-whisper for audio)	You already route uploaded images to vision models; just add a directive that picks OCR-capable models (DeepSeek OCR, PaddleOCR).
A2	Speech-to-Text (STT)	STT:	Mimo (openai-whisper), Sarvam, Meta Muse Spark	state.voiceConfig.asr = openai/whisper-large-v3 is already declared; wire the directive and re-add UI.
A3	Text-to-Speech (TTS)	TTS:	Mimo (sag ElevenLabs, sherpa-onnx-tts, mimo-tts-wav), Meta Muse Spark	state.voiceConfig.tts = nvidia/magpie-tts-multilingual is already declared; wire the directive and re-add UI.
A4	Image Generation	IMAGEN:	Meta Muse Spark (container.image.gen), Kimi (retro-tech-illustration, photo-magazine)	Browser call to a hosted image model (FLUX, SDXL, Ideogram) or your NVIDIA Integrate key.
A5	Video Generation	VIDEOGEN:	Meta Muse Spark (container.video.gen)	Same shape as A4; call Kling/Runway/Luma with async-poll for the heavy payload.
A6	Translation	TRANSLATE[from→to]:	Kimi (translation-craft), Sarvam (Indic), Meta Muse Spark	LLM shim — strict “translate only” system prompt; output is a copy-card or FILE[txt].
A7	Persistent Memory	— (system prompt references MEMORY.md)	Mimo (full MEMORY.md spec), Kimi (memory/YYYY-MM-DD.md), MiniMax	localStorage-backed; inject top entries per turn; append 1-line summary each turn.
A8	Heartbeat / Proactive Cron	— (periodic background loop)	Mimo (HEARTBEAT.md + cron)	setInterval fires a hidden prompt; model decides to email/notify or stay quiet.
A9	Humanizer	HUMANIZE:	Kimi (humanizer, humanizer-zh)	LLM shim — strict “rewrite like a human, remove AI tells” prompt; returns diff + result.
A10	Code → Diagram / Chart	CODE2DIAGRAM: / CHART:	Kimi (code-to-chart, code-to-diagram, code-arch-optimizer), Mimo (diagram-maker)	Pyodide matplotlib for charts; mermaid.js via esm.sh for diagrams, rendered in live-preview sandbox.

#	Skill	Directive shape	Source in PDF	Implementation hint
A11	Audit (schema/code/TOS)	AUDIT[type=...]:	Kimi (dataset-quality-audit, code-safety-audit, web-security-audit, tos-clause-scanner)	Pure LLM call; structured report card.
A12	PDF / File Reader	READ_PDF: / EXTRACT_TABLES:	Claude (pdf-reading, file-reading), Mimo (summarize)	Pyodide + pdfjs-dist (esm.sh) for PDF → text + rasterize pages for OCR fallback.
A13	Stock / Financial Snapshot	STOCK: / FINQUERY:	Kimi (stock-research-report, cn-finance-data, akshare, cashflow-valuation)	Browser-callable free APIs: Yahoo Finance JSON, Sina, Stooq. Cache + small table card.
A14	Cron / Scheduled Task	— (system scheduler)	Mimo (cronjob), Kimi (cronjob)	localStorage-backed scheduler; fires hidden prompt → Agent Swarm + writes to memory.
A15	R2 / Cloud Upload	UPLOAD[r2]:	Kimi (r2-upload)	Browser uploads via presigned URL from a tiny endpoint or Cloudflare R2 direct.
A16	Citation / Reference Style	CITE[apa mla chicago]:	Kimi (cite-style-converter, ref-style-converter)	Pure prompt with target-style block.

Wire-format spec — three reference skills

A6 Translation — svsystem-prompt block

TRANSLATE[from→to]: <text> # e.g. TRANSLATE[en→hi]: ...
 TRANSLATE: <text> # auto-detect source language
 Output: a copy-card with original + translated text side-by-side, plus a FILE[txt] download. Pure LLM shim: call your chat-completion model with a strict 'translate only, preserve formatting, no commentary' system prompt.

A9 Humanizer — svsystem-prompt block

HUMANIZE: <text>
 Output: original text + humanized version + a short 'changes made' diff (bullet list). Use the Wikipedia signs-of-AI-writing rubric plus a strict 'add genuine voice and personality' instruction.

A11 Audit — svsystem-prompt block

AUDIT[type=schema|dataset|code|tos]: <input>
 Output: a structured card – sections: 'Findings', 'Severity (P0/P1/P2)', 'Suggested fixes', 'References'. Pinned to one of the Kimi audit SKILL.md schemas depending on type.

4. Bucket B — Implementable but Heavier

These 12 skills fit your platform but require extra glue — usually a thin serverless relay for OAuth tokens, or they are larger integrations with non-trivial UIs.

#	Skill	Directive shape	Source in PDF	Why heavy / effort
B1	Headless Browser Automation	BROWSE_AUTO:	Mimo (agent-browser), Grok (coding-agents-harness)	Needs Node.js + Chromium sidecar — Playwright via a tiny Vercel/Cloudflare-Worker proxy.
B2	GitHub	GH[issues prs actions]: ...	Mimo (github, gh-issues), Grok (coding-agents-harness)	GitHub has a CORS-open REST API + fine-grained PAT — add GH directive, no CLI needed.
B3	Email / Calendar	EMAIL[to,subject]:	Mimo (himalaya IMAP, gog Google), Kimi (email-manager, email-newsletter-builder, email-to-calendar), Claude (morning brief)	OAuth + tiny backend to hold tokens; browser never sees the password.
B4	Notion / Obsidian / Feishu / DingTalk	NS[create read ...]: ...	Mimo (notion, obsidian, feishu-*, dingtalk-*)	Same OAuth-relay pattern as B3.
B5	YouTube Transcript	YT_TRANSCRIPT:	Mimo (video-frames, summarize)	Piped API (you already use it!) or youtube-transcript npm via esm.sh.
B6	Summarize URL / YT / Podcast	SUMMARIZE:	Mimo (summarize)	Chain FETCH: → model call → bullet card.
B7	Frontend Design / Themes	DESIGN[theme=...]:	Mimo (frontend-design, teach-impeccable, theme-factory, theme-kit)	Already covered by live_preview; bias the system prompt toward designer output.
B8	Self-improvement Log	— (per-turn)	Mimo (self-improvement, self-improving-agent)	Browser-safe: write to localStorage / IndexedDB.
B9	Tmux / Interactive Shell	—	Mimo (tmux)	Not browser-possible — skip.
B10	Multi-model Oracle (review)	ORACLE:	Mimo (oracle), Grok (nvidia-llm)	You have 15+ providers; add a directive that calls a second model for review.
B11	TDD / Code Mentor / Test-suite	MENTOR: / TDD:	Kimi (tdd-coach, test-suite-architect, code-mentor)	Pure prompt; runs in sandbox.
B12	Office QA Gates	FILE[xlsx docx pptx]: ...	Kimi (docx, Excel/XLSX, Powerpoint/PPTX) — stronger QA	Borrow Kimi's QA gates (recalc, formula validation, layout checks) and append to make_files system prompt.

5. Bucket C — Not Feasible Browser-Side

These skills require native OS access, outbound telephony, hardware bridges, or durable job-runner backends that the browser cannot host. List them in the docs so users know they are out of scope, not “missing”.

Skill / family	Reason
apple-notes / apple-reminders / bear-notes / things-mac / imsg / peekaboo	macOS-only native CLIs
tmux / camsnap / eightctl / openhue / sonoscli / spotify-player / blucli / wacli	Native CLIs and local-network control
voice-call (Twilio/Telnyx/Plivo)	Outbound telephony needs a relay
taskflow	Needs a durable detached-job backend
nano-pdf	Native CLI
coding-agent (Codex, Claude Code, OpenCode)	Each needs its own runtime

Recommendation: keep an explicit “Not available in browser” section in the docs (and a one-line tooltip in Settings → Skills) so users know these are deliberate omissions, not bugs.

6. Bucket D — Duplicates of Existing Skills

The PDF contains many skills that overlap with the 11 you already ship. Rather than add new entries, consolidate descriptions and adopt the stronger QA gates the other assistants provide.

Already in Manthan	Equivalent skill(s) in the PDF catalog
Web Search	Grok pdf · Sarvam sarvam-pdf · Kimi browse · many others
Web Scraping / Fetching	Claude pdf-reading · Mimo smart-web-scraper · Kimi batch-download
Code Interpreter	Kimi code-arch-optimizer · Kimi deep-module-refactor
File Creation	All 13 sources ship a docx/pdf/xlsx/pptx variant
Live Preview	Grok live-preview · Mimo frontend-design · Kimi webapp-building · webapp-building-swarm
Skill Creator	Mimo skill-creator · Grok skill-creator · Claude skill-creator
MCP Date	— (no duplicates)
MCP Calculator	— (no duplicates)
Agent Swarm	Mimo taskflow · Kimi vibecoding-general-swarm · webapp-building-swarm · backend-building-swarm
Web Browsing	Kimi browse · Mimo agent-browser (heavier alternative)

Action: keep one canonical entry per skill in SAHU_SKILLS; where the PDF source has a stronger QA gate or richer prompt, port the relevant paragraphs into the existing skill's `buildSearchLoopPrompt()` block.

7. Implementation Roadmap

The 16 Bucket-A skills split naturally into four sprints ordered by effort × user-visible value.

Sprint 1 — Quick wins (1–2 days)

- A6 Translation
- A9 Humanizer
- A11 Audit (schema / dataset / code / TOS)
- B6 Summarize (URL / YT / Podcast / PDF)
- B7 Frontend-Design / Theme bias
- B11 TDD / Code Mentor / Test-suite

Sprint 2 — Wire up dormant voice layer (1 day)

- A2 STT (re-add mic UI, hook whisper-large-v3)
- A3 TTS (re-add audio-out, hook magpie-tts-multilingual)

Sprint 3 — Vision & media (2–3 days)

- A1 OCR (DeepSeek OCR / PaddleOCR directive)
- A4 Image Generation (FLUX / SDXL / Ideogram)
- A5 Video Generation (Kling / Runway / Luma)

Sprint 4 — Long-running memory (1–2 days)

- A7 Persistent Memory (MEMORY.md in localStorage)
- A8 Heartbeat / Proactive Cron
- A14 Cron / Scheduled Tasks
- B8 Self-improvement log

Sprint 5 — Bigger integrations (1–2 weeks)

- B2 GitHub (PAT + REST, no CLI needed)
- B3 Email / Calendar (OAuth relay)
- B4 Notion / Obsidian / Feishu / DingTalk
- B5 YouTube Transcript
- B10 Multi-model Oracle

Sprint 6 — Backend-required (optional)

- B1 Headless browser automation (Playwright proxy)
- B12 Office QA gates from Kimi

8. Wire-Format Specification (Sprint 1 Reference)

For each new skill you need four pieces in `index_2.txt`:

(i) an entry in the SAHU SKILLS array

```
{
  id: 'translate',
  name: 'Translation',
  icon: 'fa-language',
  color: 'teal',
  desc: 'Translate text between languages with a strict LLM shim '
    + '(auto-detect or specify from→to). Outputs original + '
    + 'translated side-by-side.',
},
```

(ii) a svstem-prompt branch in buildSearchLoopPrompt()

```
if (has('translate')) {
  lines.push(
    'TRANSLATE[from→to]: <text> // e.g. TRANSLATE[en→hi]: ...\\n'
    + 'TRANSLATE: <text> // auto-detect source\\n'
    + 'Output: a copy-card with original + translated text side-by-side, '
    + 'plus a FILE[txt] download.'
  );
}
```

(iii) a directive extractor (regex + handler)

```
// In the directive-parser block:
const TRANSLATE_RE = /TRANSLATE(?:\[([^\]]+\)])??:\s*([^\n]+)/gi;
// Match: TRANSLATE[en→hi]: Hello world
// Match: TRANSLATE: Bonjour le monde

function handleTranslate(text, lang) {
  const result = callChatCompletion({
    systemPrompt: 'You are a translator. Output ONLY the translated '
    + 'text, preserving formatting. No commentary.',
    userMessage: (lang ? 'Translate from ' + lang + '.\\n\\n' : '')
    + text,
  });
  return renderCopyCard({ original: text, translated: result });
}
```

(iv) a card renderer for the user-facing bubble

```
function renderCopyCard({ original, translated }) {  
  return '<div class="translate-card">'  
  + '<div class="translate-original">'  
  + '<span class="label">Original</span>'  
  + '<div class="text">' + escapeHtml(original) + '</div>'  
  + '</div>'  
  + '<div class="translate-arrow"><i class="fa fa-arrow-right"></i></div>'  
  + '<div class="translate-result">'  
  + '<span class="label">Translation</span>'  
  + '<div class="text">' + escapeHtml(translated) + '</div>'  
  + '<button class="btn btn-ghost btn-icon copy-btn">'  
  + '<i class="fa fa-copy"></i></button>'  
  + '</div>'  
  + '</div>';  
}
```

9. Source Catalog Summary

All recommendations in this document were derived from the PDF catalog's 13 sources. Their sizes and skill counts (as of 25 September 2026):

#	Source	Pages	Skills	Notes
1	Sarvam AI	2	6	Tightly scoped bound skill packs
2	ChatGPT (OpenAI)	4	13	Tool-family inventory
3	Grok (xAI)	5	9	SKILL.md files
4	GLM (Z.ai)	14	12	Capability handbook
5	Mimo (Xiaomi, OpenClaw)	81	103	Full skill atlas + SOUL.md + AGENTS.md
6	MiniMax (MAVIS)	42	17	Reference doc — this is Mavis
7	DeepSeek	—	~50	Source as Python script
8	hy (research agent)	—	2	hy_search + hy_visit
9	Claude (Anthropic)	3	11	SKILL.md collection
10	Gemini (Google)	—	—	Skill matrix by domain
11	Meta AI (Muse Spark)	—	—	Capability catalog + tools
12	Kimi (Moonshot AI)	1,073	271	Full SKILL.md collection
13	Manthan Agent (SahuAI)	—	11	Subject of this document

Total: **13 assistants, well over 400 distinct skills**. Of these, 16 Bucket-A skills + 12 Bucket-B skills = 28 are implementable in Manthan Agent; the rest are duplicates, browser-infeasible, or already covered by your current 11.

Companion documents

- Unified_Skill_Report.pdf / .md — condensed cross-assistant capability matrix.
- Unified_Bug_Report.pdf / .md — current bug register (consumed by the v7.9.13 fix pass).
- index_2.txt — the Manthan Agent v7.9.12 source file (subject of this plan).
- manthan-skill-expansion-plan.pdf — this document.

End of document. For questions or to extend the plan (e.g. wire-format for A1 OCR, A4 Image Generation, or B10 Oracle) please reply with the target section number and the desired output format.